

PROJECT TOPIC: SURVEY OF BIODIVERSITY AND HERITAGE TREES

Topic Description: Survey of Urban and Village Biodiversity Focusing on Centuries-Old Heritage Trees in Maharashtra: Ecological Value, Wildlife Support, and Legal Protection

Trees are the longest-living biological organisms on our planet, serving as living botanical monuments that bridge centuries of ecological, cultural, and landscape evolution. In an era of breakneck urbanization, infrastructure expansion, and highway widening, ancient veteran trees are routinely chopped down in minutes, erasing centuries of microclimate moderation, carbon storage, and wildlife habitats that cannot be replaced by planting a handful of fragile saplings.

This project documents a systematic biodiversity survey and field inventory of **Heritage Trees (हेरिटेज वृक्ष)** in our local urban/village landscape in Maharashtra, conducted in light of the landmark *Maharashtra (Urban Areas) Protection and Preservation of Trees (Amendment) Act, 2021*, which statutorily defines a 'Heritage Tree' as any tree possessing an estimated age of 50 years or more.

The survey specifically locates, measures, documents, and evaluates iconic veteran trees—primarily mature **Banyan** (*Ficus benghalensis*), **Peepal** (*Ficus religiosa*), **Tamarind / Chinch** (*Tamarindus indica*), and **Neem** (*Azadirachta indica*). By recording their tree girth, canopy spread, nesting avifauna, and cultural significance, the project demonstrates why heritage trees are irreplaceable biological keystones.

2. Importance of the topic (विषयाचे महत्त्व)

Heritage trees perform multifaceted ecological, historical, and microclimatic functions that young saplings cannot replicate for decades:

Ecological Supremacy of Heritage Trees :-

- **Massive Carbon Storage & Oxygen Production:** A single 100-year-old mature Banyan tree sequesters several tons of carbon dioxide in its massive woody trunk and root biomass, while releasing enough daily oxygen to support dozens of human beings.
- **Complex Vertical Micro-Ecosystems:** A veteran tree is not a single plant; it is an entire vertical ecosystem. Its deeply furrowed bark, hollow cavities, and spreading branches shelter hundreds of insect species, epiphytic orchids, bats, owls, squirrels, and nesting raptors.
- **Microclimate & Urban Heat Island Mitigation:** The expansive 40-meter canopy of a heritage tree provides dense shade, reducing ambient surface asphalt temperatures beneath it by 6°C to 10°C through intense evapotranspiration.
- **Groundwater Anchors & Deep Aquifer Channels:** Deep, extensive century-old root networks create natural macro-pores in the subsoil, channeling monsoon rainwater deep into underground aquifers and preventing urban street flooding.

Felling a heritage tree destroys an entire biological civilization that took centuries to build.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The field survey was designed around structured arboricultural, botanical, and legal protection objectives:

Structured Project Objectives :-

- **Field Location & GPS Geo-Tagging:** To survey, locate, photograph, and record GPS coordinates for four prominent heritage trees (estimated age > 50 years) in the local area.
- **Dendrometric Measurements:** To measure tree girth at breast height (GBH), estimate total canopy crown diameter, and determine approximate tree age using forestry standard formula.
- **Associated Biodiversity Audit:** To observe, identify, and catalog all birds, arboreal mammals, reptiles, and insects utilizing each heritage tree for roosting, nesting, or feeding.
- **Threat & Vulnerability Assessment:** To document anthropogenic threats—including root-zone asphalt paving, billboard nailing, trash burning, and road-widening proposals.
- **Legal Protection Action Plan:** To utilize the provisions of the Maharashtra Tree Amendment Act of 2021 to formally advocate for the heritage protection of surveyed trees.

The scope encompasses physical measuring tapes, binoculars, photographic logs, and community interviews.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized dendrometric, arboricultural, and ecological survey methodology was deployed:

Methodological Survey Steps :-

- **Phase 1 – Field Transect Reconnaissance:** Walked a 2-kilometer survey transect across historic village/suburban nodes (temple courtyards, village squares, historic avenues) to identify candidate heritage trees.
- **Phase 2 – Girth & Canopy Measurement:** Measured Girth at Breast Height (GBH at 1.3 meters above ground) using a flexible measuring tape; estimated crown diameter by stepping off canopy drip-lines.
- **Phase 3 – Age Estimation Protocol:** Applied the standard forestry formula:
Estimated Age (Years) = Girth (cm) / Annual Growth Factor, cross-verifying results through historical community records and oral interviews with village elders.
- **Phase 4 – Wildlife & Cavity Survey:** Spent 30 minutes at each tree during early morning hours using binoculars to record nesting holes, bird species, and insect colonies.
- **Phase 5 – Report Synthesis:** Compiled findings into structured matrices and drafted civic preservation proposals under the Maharashtra Tree Act.

All measurements were recorded safely on the ground without damaging tree bark or disturbing nesting wildlife.

5. Observations (निरीक्षणे)

Systematic survey across the study transect documented four majestic heritage trees supporting rich urban biodiversity:

5.1 Tabular Record of Four Surveyed Heritage Trees

Species & Location	Girth & Est. Age	Observed Wildlife & Nesting	Observed Threats & Physical Condition
Banyan (Vad) <i>Ficus benghalensis</i> (Near Village Chavdi)	Girth: 6.8 meters Crown: 38 meters Est. Age: ~120 Yrs	Colony of 40+ Indian Flying Foxes (fruit bats); 6 active nesting pairs of Rose-ringed Parakeets; Asian Koel; honeybee combs.	Road widening project poses imminent threat; cement paver blocks laid right touching prop roots; shop banners nailed to trunk.
Peepal (Pimpal) <i>Ficus religiosa</i> (Historic Temple Square)	Girth: 5.2 meters Crown: 26 meters Est. Age: ~95 Yrs	Two nesting cavities of Spotted Owlets; Coppersmith Barbets feeding on figs; thousands of black ants and carpenter bees.	Concrete platform (Chabutara) built tightly choking root collar; oil and wax lamps burned against bark during festivals.
Tamarind (Chinch) <i>Tamarindus indica</i> (Old District Highway)	Girth: 4.4 meters Crown: 22 meters Est. Age: ~85 Yrs	Roosting site for Common Mynas, Black Kites, and Tree Pies; Palm Squirrels caching seeds in deep bark fissures.	Base used as an unauthorized garbage dump; periodic trash burning has severely scorched the lower trunk; highway expansion risk.
Neem (Kadu-Neem) <i>Azadirachta indica</i> (Old School Campus)	Girth: 3.6 meters Crown: 18 meters Est. Age: ~65 Yrs	Dense nesting colonies of Brahminy Starlings and Red-vented Bulbuls; rich hoverfly and pollinating insect diversity.	Severe unscientific branch lopping by electricity board contractors; heavy pruning has destabilized canopy balance.

6. Analysis of data (माहितीचे विश्लेषण)

Analytical examination of the survey data highlights the irreplaceable ecological supremacy of veteran trees and their extreme urban vulnerability:

Core Analytical Inferences :-

- **Compensatory Afforestation is a False Equivalent:** Modern infrastructure developers claim that chopping a 100-year-old Banyan tree can be compensated by planting 10 small saplings elsewhere. Ecological science proves this is false: a sapling takes 50 to 80 years to develop the canopy volume, trunk cavities, and wildlife-carrying capacity of a single heritage tree.
- **The Choking Threat of Concretization:** All four surveyed trees suffered from root-collar suffocating: municipal asphalt and interlocking paver blocks sealed the soil surface, blocking rainwater infiltration and root gas exchange. This causes internal root rot and leads to sudden tree falls during squalls.
- **Hollow Cavities as Crucial Wildlife Niches:** Young plantation trees have smooth, solid wood with zero cavities. Only veteran heritage trees develop the natural hollows and dead branches essential for hole-nesting birds like Spotted Owlets, Parakeets, and Barbets.
- **The 2021 Maharashtra Tree Act as a Legal Weapon:** The survey proves that citizens can use the amended Tree Act to protect these trees. The law mandates that anyone seeking to fell a Heritage Tree must plant compensatory trees equal to the age of the felled tree, creating a strong economic deterrent against cutting.

The analysis affirms that preserving existing heritage trees must take absolute precedence over planting new saplings.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Heritage trees are living national monuments that anchor urban biodiversity, cool our cities, and preserve historical memory. Protecting these ancient giants against mindless road widening and root concreting is an urgent civic, legal, and environmental duty for all citizens.

Action Checklist for Heritage Tree Conservation :-

- ❑ Submit survey records to the local municipal Tree Authority to officially register trees as 'Heritage Trees'.
- ❑ Physically de-concretize a 1-meter radius around the base of mature avenue trees to allow root breathing.
- ❑ Remove all iron nails, advertising banners, cables, and barbed wire attached to ancient tree trunks.
- ❑ Stop the burning of municipal sweepings and dry garbage heaps near the bases of avenue trees.
- ❑ Advocate for engineering highway curves around heritage trees rather than felling them for straight roads.

8. Bibliography / References (संदर्भ सूची)

1. Government of Maharashtra. (2021). *The Maharashtra (Urban Areas) Protection and Preservation of Trees (Amendment) Act*.
2. Randhawa, M. S. (1957). *Flowering Trees in India*. Indian Council of Agricultural Research, New Delhi.
3. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.
4. Thomas, Peter. (2014). *Trees: Their Natural History (2nd Edition)*. Cambridge University Press.